RELATIVE MAYER-VIETORIS DISTINGUISHED TRIANGLE FOR THE DERIVED PUSHFORWARD IN THE UNBOUNDED DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the functoriality reference; no step is referred to the reader.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-unbounded-relative-mayer-vietoris is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- K-injective resolutions with injective terms exist, functorially. Injectives, Theorem injectives-theorem-K-injective-embedding-grothendieck, invoked at proof lines 8267-8271 and again at 8292-8294. Supplies the representative and the functoriality of the construction.
- Restriction to an open subspace preserves K-injectivity. Same chapter, Lemma lemma-restrict-K-injective-to-open, invoked at proof lines 8273-8274. Supplies that the restrictions compute the derived pushforwards along the three restrictions of f.
- Relative Mayer-Vietoris and the vanishing of R^1 f_* of an injective module. Same chapter, Lemma lemma-relative-mayer-vietoris, invoked at proof lines 8288-8291. Supplies exactness of the sequence of complexes of sheaves on Y.
- Distinguished triangle attached to a short exact sequence of complexes. Derived Categories, Section derived-section-canonical-delta-functor and Lemma derived-lemma-derived-canonical-delta-functor, invoked at proof lines 8276-8280. Supplies the triangle once exactness is known.

Source reference. Original lemma 0152: sources/stacks-chow/tags.txt, source lines 1493–1493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 014Z: sources/stacks-chow/tags.txt, source lines 1490–1490. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original theorem 079P: sources/stacks-chow/tags.txt, source lines 9279–9279. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EC: sources/stacks-cohomology/pinned-original-source.tex, source lines 804–804. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08BS: sources/stacks-cohomology/pinned-original-source.tex, source lines 7840–7840. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of ringed spaces. Suppose that $X = U \cup V$ is a union of two open subsets. Denote $a = f|_U : U \to Y$, $b = f|_V : V \to Y$, and $c = f|_{U \cap V} : U \cap V \to Y$. For every object E of $D(\mathcal{O}_X)$ there exists a distinguished triangle*

$$Rf_*E \to Ra_*(E|_U) \oplus Rb_*(E|_V) \to Rc_*(E|_{U \cap V}) \to Rf_*E[1]$$

This triangle is functorial in E.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ representing E. We may assume $\mathcal{I}^n$ is an injective object of $Mod(\mathcal{O}_X)$ for all n, see Injectives, Theorem 079P. Then Rf_*E is computed by $f_*\mathcal{I}^\bullet$. Similarly for U, V, and $U \cap V$ by Lemma 08BS. Hence the distinguished triangle of the lemma is the distinguished triangle associated (by Derived Categories, Section 014Z and especially Lemma 0152) to the short exact sequence of complexes

$$0 \to f_*\mathcal{I}^\bullet \to a_*\mathcal{I}^\bullet|_U \oplus b_*\mathcal{I}^\bullet|_V \to c_*\mathcal{I}^\bullet|_{U \cap V} \to 0.$$

This is a short exact sequence of complexes by Lemma 01EC and the fact that $R^1 f_*\mathcal{I} = 0$ for an injective object $\mathcal{I}$ of $Mod(\mathcal{O}_X)$. The final statement follows from the functoriality of the construction in Injectives, Theorem 079P. □